Task 2 - Closed Knight's Tour on an $n \times n$ Board

Problem Description

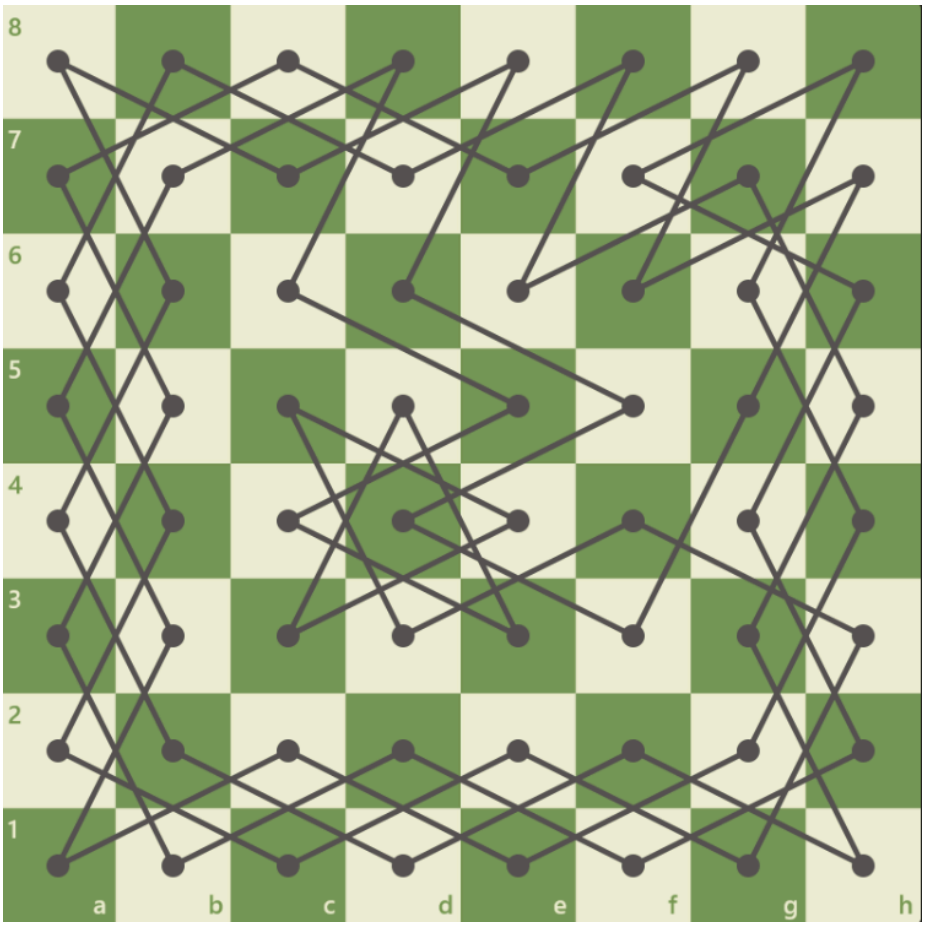
Is it possible for a chess knight to visit all the cells of an 8×8 chessboard exactly once, ending at a cell one knight's move away from the starting cell? (Such a tour is called closed or re-entrant. Note that a cell is considered visited only when the knight lands on it, not just passes over it on its move.)
What if the chessboard is $n \times n$? is all n where n>8 is possible for a chess knight to visit all the cells? If not, what is the cases?
design a greedy algorithm to find the minimum number of moves the chess knight needs. And study the different cases of n

Problem Analysis and Understanding

Subtask Breakdown

Subtask 1: is the Knight's Tour Possible on an 8x8 chessboard?

Yes. The Closed Knight's tour is a very famous problem that has been researched by mathematicians for centuries, including Euler himself. One of the earliest solutions to the problem is the one shown here:



it was actually discovered later on that there are 26,534,728,821,064 possible closed tours.

Subtask 2: For Which n x n Boards Does a Closed Tour Exist?

Key Results:

n	Closed Tour?	Reason
1	No	Only 1 cell, no moves possible
2	No	Board too small, knight cannot reach all cells
3	No	Board too small, knight gets stuck
4	No	Proven impossible (exhaustive search)
5	No	n is odd -> impossible (coloring argument)
6	Yes	First board size where closed tour is possible
7	No	n is odd -> impossible
8	Yes	Classic problem, many solutions exist
n >= 6, even	Yes	Constructive proofs exist
n odd	**Never**	Impossible by the coloring argument

Conclusion: A closed knight's tour on an n x n board exists **if and only if n is even and n >= 6**.

Clarification: Why Odd n's are Always Impossible

Color the board like a standard chessboard (alternating black and white). A knight **always** moves from a black square to a white square and vice versa. Therefore, if a board is for example of size 7x7, then there'll be an unequal number of black and white tiles, causing it to be impossible to traverse the entire board without repeating a square.

Formal Proof

1. A knight's tour on n x n visits n^2 squares.
2. A closed tour forms a **Hamiltonian cycle** on the knight graph.
   a graph cycle where each node is visited exactly once before returning to the starting node.
3. The knight graph is **bipartite** (black squares vs. white squares).
   a bipartite graph is a graph where nodes are split into two separate sets, with no edges connecting two nodes within the same set.
4. A Hamiltonian cycle in a bipartite graph requires **equal-sized** sets.
5. When n is odd, n^2 is odd, so one color has (n^2+1)/2 squares and the other has (n^2-1)/2 squares -- **unequal**.
6. Therefore, **no closed tour exists for any odd n**.

Subtask 5: Designing a Greedy Algorithm for this problem

Goal: Design a greedy algorithm to find a closed knight's tour.

Approach: Use **Warnsdorff's Algorithm** (1823), a classic greedy heuristic:

- From the current square, move to the neighbor that has the **fewest onward moves** (i.e., the fewest unvisited neighbors).
- Ties are broken by preferring squares closer to the board center (or by trying all tied options).

For **closed** tours specifically, we must also ensure the last square is a knight's move away from the starting square. This may require backtracking on the final steps.

References

Knight's Tour Explanation: https://www.chess.com/terms/knights-tour-chess

another source: https://www.youtube.com/watch?v=ab_dY3dZFHM

WarnsDorffs Algorithm: https://www.geeksforgeeks.org/dsa/warnsdorffs-algorithm-knights-tour-problem/